

Assignment 2

Q1 – CLO 1

Scenario:

You have trained a Logistic Regression model to detect fraudulent transactions. Your test set contains 10,000 transactions, out of which only 50 are actual frauds. Your model predicts that every transaction is legitimate (non-fraudulent).

1. Construct the Confusion Matrix for this outcome. Clearly label all four cells — TP, FP, FN, and TN — with their numeric values. [2 marks]
2. Calculate the Accuracy and the Recall of this model. Then explain, in two to three sentences, why Accuracy is a fundamentally flawed metric for this dataset even though it appears high. [4 marks]
3. A colleague proposes using SMOTE to address the class imbalance. Explain the geometric mechanism of SMOTE — specifically how it generates new synthetic minority-class samples using nearest neighbours in feature space. Then contrast SMOTE with Cost-Sensitive Learning, which modifies the loss function rather than the data. For each approach, state one practical advantage and one limitation. [4 marks]

Q2 – CLO 2

Your dataset contains two clusters shaped as concentric rings — an inner ring of Class A surrounded by an outer ring of Class B.

1. Explain why K-Means will fail to separate these rings, referencing its objective function. [5 marks]
2. A teammate argues for Agglomerative Clustering with complete-linkage; you argue for single-linkage. Who is correct for this geometry, and why? Define both linkage criteria precisely and explain how complete-linkage's definition causes it to fail on elongated, ring-shaped structures. [5 marks]

Q3 – CLO 3

Scenario:

You are building a model to predict student exam scores based on features like study hours, attendance rate, and department (a categorical variable). Your preprocessing involves a StandardScaler for numerical features and a TargetEncoder for the categorical feature. A junior teammate proposes the following order: fit both transformers on the full dataset → split into train/test → train the model.

1. Identify and name the specific leakage introduced. [5 marks]
2. Redraw the correct pipeline order, clearly marking at which step each transformer is fitted vs. only applied. [5 marks]

3. Your teammate argues: 'the test set is only 20% of data, so the leakage effect is negligible.' Refute this argument with a concrete reasoning. [5 marks]

Q4 – CLO 4

A deployed model predicting daily ride-sharing demand performed excellently for the first three months. By month six, the Mean Absolute Error (MAE) has tripled, despite the API and servers functioning perfectly.

1. Differentiate between Data Drift and Concept Drift. For each type, provide one plausible real-world example that could specifically cause this ride-sharing demand model to degrade in the manner described. [5 marks]
2. Explain how you would implement a Kolmogorov–Smirnov (KS) Test check within your production logging pipeline to detect feature drift before the model's accuracy fully degrades. Your answer should describe: (a) what two distributions the test compares, (b) what the test statistic measures, and (c) how a significant result ($p < 0.05$) would trigger an engineering response. [5 marks]
3. Describe one method of Red Teaming this model before its initial deployment. Then define what constitutes an Adversarial Input in the specific context of tabular regression data — contrast this with the more commonly discussed image-domain adversarial examples to highlight the key differences. [5 marks]